

```
public static int task5CM(String input1, String input2) {  
    int m = input1.length();  
    int n = input2.length();  
    int[][] dp = new int[m + 1][n + 1];  
    for (int i = 1; i <= m; i++) {  
        for (int j = 1; j <= n; j++) {  
            if (input1.charAt(i - 1) == input2.charAt(j - 1)) {  
                dp[i][j] = dp[i - 1][j - 1] + 1;  
            } else {  
                dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);  
            }  
        }  
    }  
    return dp[m][n];  
}
```

```
public static void main(String[] args) {  
    String input1 = "abcde";  
    String input2 = "ace";  
    int result = task5CM(input1, input2);  
    System.out.println(result);  
}
```